

Release Gates for Physical AI

A Unified Framework for Functional, Adversarial, Cyber, and Hardware
Verification of Learned Robot Systems

Shubh Jain

Carnegie Mellon University

shubhj@andrew.cmu.edu

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Status. This is a framework and position paper. It contains no new experimental results. Every quantitative figure is attributed to published work or to a named standard; every threshold proposed here is labelled as a proposal and is not yet validated. **Scope.** Manipulation and station-based mobile manipulation. Locomotion, whole-body balance, and public-road autonomy are out of scope.

Abstract

Learned manipulation policies are now updated on the cadence of software, shipped to fleets monthly, and evaluated with the rigour of a demo. Classical machine-safety verification assumes a deterministic controller whose behaviour can be argued from its structure; a vision-language-action policy offers neither determinism nor structure, and its competence is a property of a distribution rather than of a program. The decision to deploy a new policy version is consequently made today without a coherent body of evidence.

We observe that the evidence bearing on that decision already exists in four separate literatures and four separate industrial practices—functional evaluation, adversarial robustness, product cybersecurity, and hardware reliability—and that no framework composes them. We propose one. Our contributions are: (i) a taxonomy of the four test families that bear on a release decision, with the metrics each can produce; (ii) a *fidelity-gating* principle, under which a test may block a release only once its predictive validity against real outcomes has been measured, and otherwise reports as advisory; (iii) a release-scoped, immutable evidence model mapped to the clauses of ISO 10218:2025 and the EU Cyber Resilience Act, including the treatment of model weights as components of a software bill of materials; and (iv) an evaluation protocol for the framework itself, centred on a retrospective study that asks whether a proposed gate would have caught regressions that actually reached the field.

We further argue that the composition is not merely administrative: several failure modes are visible only across families—a policy version that improves task success while becoming more susceptible to in-scene prompt injection, or whose grasp failures concentrate on units whose end-effectors are past a wear threshold—and are structurally invisible to any single-family evaluation.

1 Introduction

A robot deployment running a learned policy carries two independent kinds of uncertainty: whether the policy will do the task, and whether the surrounding system will behave when something adversarial, degraded, or unforeseen occurs. Industrial practice has mature answers for neither, and no answer at all for their interaction.

The immediate cause is cadence. A fleet running a vision-language-action (VLA) policy receives new weights monthly or faster. The governing safety standard for industrial robot applications was revised in 2025 to accommodate, among other things, machinery incorporating machine-learning functions that alter behaviour over time [, b,c], but conformity assessment remains an event rather than a subscription. Between two certified states a policy may change behaviour substantially, and nothing in the classical process observes that change.

The deeper cause is epistemic. Functional safety argues from structure: a speed-and-separation monitor can be shown to be correct. A learned policy has no structure to argue from, so its competence must be *measured*, on a distribution, repeatedly. Measurement of that kind produces evidence with an unfamiliar property: it is only as good as its own predictive validity. A simulated benchmark that does not rank policy versions the way reality ranks them is not weak evidence but misleading evidence, and a release process that treats the two alike is worse than one with no simulation at all.

1.1 The four silos

Ask a robot company for the evidence behind its last release and four disconnected artifacts arrive: a notebook of task success rates on held-out cells; a penetration-test report, usually more than a year old, that does not mention the policy; maintenance records in a separate system describing hardware whose degradation nobody has related to task performance; and a safety file at a notified body describing a configuration that predates the current weights.

Each silo has a serious literature. Real-to-sim evaluation now reports rank correlations against real-robot outcomes of 0.911 [sim, 2026] and 0.887 [rob, 2026c]. Physical prompt injection has a systematic taxonomy and a 5,670-trial benchmark [pap, 2026]. Product cybersecurity has a regulation with dated obligations and defined artifacts [, a]. Hardware reliability has an established engineering practice. What is missing is the composition, and the composition is where the release decision lives.

1.2 Contributions

1. A **test taxonomy for the release decision** (§4.1): four families, their runners, and the metrics each can produce with current published methods.
2. **Fidelity gating** (§4.3): a rule separating tests permitted to block a release from tests permitted only to advise, based on measured predictive validity rather than on apparent realism.
3. A **release-scoped evidence model** (§4.4–§4.5) with an explicit mapping to ISO 10218:2025 and CRA artifacts, including model weights as SBOM components.
4. An **evaluation protocol for the framework** (§6), centred on a retrospective study of known field regressions.
5. A statement of **open problems** (§7), the most consequential being that the verifiers this framework depends on are, on contact-rich tasks, close to unusable today [fai, 2026].

2 Related work

2.1 Functional evaluation

Evaluating a policy in simulation against real outcomes has moved from aspiration to measurement. SimplerEnv established visual-matching and variant-aggregation protocols for comparing simulated and real manipulation performance [sim, 2024]. Recent automated real-to-sim systems report correlation directly: SimFoundry builds simulation-ready scenes from RGB video and reports a mean Pearson correlation of 0.911 with a maximum rank violation of 0.018 across seven tasks and five policy types, at roughly five minutes of reconstruction per object, with F_1 between 0.81 and 0.92 fully automated and 0.93–0.99 given about three minutes of per-object human refinement [sim, 2026]. RoboSnap reconstructs from a single image and reports a Pearson correlation of 0.887 and successful replay of five of five real trajectories [rob, 2026c]. Both are explicit that physical parameters—mass, friction—are assigned by querying a vision-language model rather than measured; RoboSnap states that it includes no dedicated pipeline for automatic physical-parameter estimation [rob, 2026c], and an earlier framework attributes its residual error to “domain mismatch, mesh reconstruction errors, and discrepancies in material property estimates” [sca, 2025].

A second line replaces the simulator with a learned world model and scores policies inside it [one, 2025, gig, 2026, dwo, 2026]. The trade is legibility: a learned evaluator yields no contact forces, no queryable state, and no counterfactual outside its training distribution, which makes it unsuitable as the substrate for cyber or hardware evidence even where its functional predictions are good.

Underneath both sits an unresolved dependency: deciding whether an episode succeeded. FailBench evaluates thirteen VLM-based success detectors and finds the best reaching 0.77 mean balanced accuracy, degrading to below 0.60—near chance—on contact-intensive assembly, with a systematic bias toward predicting success under ambiguous evidence; models fine-tuned for failure detection underperformed general-purpose VLMs [fai, 2026]. A companion survey catalogues verifier designs and their failure modes [nof, 2026], and calibrated task-success confidence is an active target [vla, 2026a].

2.2 Adversarial robustness

Embodiment changes the adversarial threat model, because the consequence is physical and frequently irreversible [vla, 2026b]. The sharpest current result is physical prompt injection: adversarial text placed in the robot’s visual field acts as an indirect injection into a VLM planner’s reasoning. Across 5,670 trials with a four-category taxonomy—indirect signage, task redefinition, authority impersonation, conflict injection—attack success reached 27.0%, 29.4%, and 5.0% on three frontier models [pap, 2026]. Related work demonstrates trajectory-level redirection, in which the instruction still appears to specify the intended task while the physical outcome is changed [tra, 2026], partially observable adversarial patches [pat, 2026], and interference through physical signals such as sound, light, and electromagnetic emission.

Defensively, adversarial scenario generation is being formulated as a red-team/blue-team game [adv, 2026], boundary-focused sampling is used for failure discovery before deployment [rob, 2026b], and the formal-methods community has surveyed falsification for learned policies [for, 2026a]. None of this work is packaged as release evidence.

2.3 Cybersecurity and standards

Two instruments set dates. The revised ISO 10218-1 and 10218-2, published in 2025, absorb collaborative requirements formerly in ISO/TS 15066 and add robot classifications with matching functional-safety requirements, safety-related cybersecurity requirements, and end-effector guidance [b,c]. ISO 25785-1, the first standard written for dynamically stable industrial mobile robots—the humanoids and quadrupeds now entering warehouses—remained a committee draft in early 2026 [d]. Separately, the EU Cyber Resilience Act entered into force in December 2024; vulnerability and incident reporting obligations apply from 11 September 2026, and full compliance—CE marking, machine-readable SBOM, secure-by-design requirements, vulnerability handling, and security updates across the support period—from 11 December 2027, with penalties up to EUR 15M or 2.5% of global turnover [a, cra, 2026].

Vendor safety stacks have moved faster than the standards. Halos for Robotics, announced in June 2026, couples safety compute, sensor bridging, a safety OS, and an accredited inspection lab that prepares integrations for third-party certification against IEC 61508, ISO 13849, and ISO/IEC TR 5469 [NVIDIA, 2026]; FORT Robotics extends it with infrastructure-side sensing [for, 2026b]. This is functional safety: it bounds harm to people, does not measure task competence, and does not address adversarial manipulation of the policy’s inputs. Control-barrier formulations that embed ISO 10218 constraints directly in the controller [cbf, 2026] share that boundary.

2.4 Hardware reliability and its coupling to policy

Industrial practitioners consistently identify actuation as the binding constraint: a full-body platform carries thirty or more degrees of freedom, each required to deliver repeatable torque across thousands of hours, and harmonic-drive flex-splines are subject to fatigue under exactly the cyclic loading that walking and repetitive assembly produce; end-effectors are reported as the most failure-prone and most expensive subsystem [hum, 2026b,a]. Absent from the literature is the coupling: no published framework relates a measured degradation state—a suction cup past a wear threshold, a drifting camera extrinsic—to the task success of the specific policy version running on that unit. In deployment this coupling is routine, and it is the most common informal explanation for “the same policy performs differently across sites.”

2.5 What no existing system composes

Independent evaluation infrastructure is emerging—community and hosted real-robot benchmarks [rob, 2026a, 2025], low-cost reproducible rigs [pha, 2026, rep, 2026]—and is converging on public leaderboards for functional performance. Industrial V&V platforms in the adjacent automotive domain integrate reconstruction, scenario generation, and regulatory-grade reporting, with a published three-tier metric framework separating reconstruction fidelity, open-loop stack agreement, and closed-loop behavioural comparison against real drive data [Applied Intuition, 2026a,b]. We take that three-tier structure as prior art and generalise it, noting that its robotics analogue must additionally carry cyber and hardware evidence, because in manipulation the same release decision is gated on all four.

3 Problem statement

Definition 1 (System under test). A system under test is the tuple

$$S = (h, \pi, \sigma, c) \quad (1)$$

with h the hardware configuration (kinematics, end-effector, sensors, and their measured degradation state), π the policy checkpoint, σ the software and firmware stack including model dependencies, and c the site configuration (fixtures, item catalogue, lighting, network posture). A release changes at least one component; evidence is valid only for the tuple it was produced against.

Definition 2 (Evidence). An evidence record is

$$e = (t, r, S, y, \rho, p) \quad (2)$$

for test specification t , runner r , system under test S , outcome y , fidelity estimate ρ for the pair (t, r) , and provenance p (inputs, versions, seeds, operator). Records are immutable and scoped to a release.

Definition 3 (Gate). For a candidate S' and baseline S , a family of tests partitions into a gating set G and an advisory set A by

$$t \in G \iff \rho(t, r) \geq \tau \wedge n(t) \geq n_{\min}. \quad (3)$$

The release is blocked if any $t \in G$ shows a regression beyond its tolerance. Tests in A are reported with their fidelity and cannot block. We propose $\tau = 0.8$ as a rank-correlation threshold (§5), by analogy with the 0.887–0.911 correlations reported by current real-to-sim systems [sim, 2026, rob, 2026c]; the value requires empirical justification per domain.

Three consequences follow. First, **fidelity is a property of a test–runner pair, not of a simulator**: the same engine may be gating-grade for a rigid pick and advisory-grade for cable routing. Second, **evidence expires** when any component of S changes, which is why hardware degradation belongs inside the model rather than beside it. Third, **a release decision is a multiple-comparison problem** across tasks and families, and must be treated as such rather than as a single averaged score.

4 Framework

4.1 Test families and their metrics

Table 1 states what each family tests, where it can run today, and what it produces. The *status* column records our assessment of whether current published methods support gating-grade evidence, and is the honest content of this paper: two of four families cannot presently gate on contact-rich work.

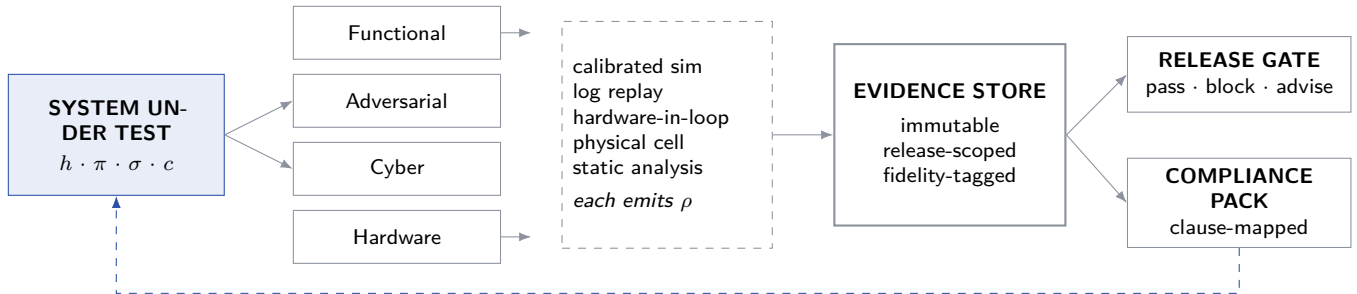


Figure 1: The release-gate pipeline. Four test families execute on five runner types; every result carries a fidelity estimate that determines whether it may block a release. The evidence store is the system of record from which both the gate decision and the clause-mapped compliance pack are derived. Field failures and degradation telemetry re-enter as new test specifications, scoped to the tuple on which they were observed.

4.2 Runners

A runner is any mechanism that can execute a test specification and return an outcome: a calibrated simulator, replay of logged episodes against a policy, hardware-in-the-loop, a physical cell, or static analysis of the software stack. Runners are interchangeable within a family and are the unit at which fidelity is estimated. The framework deliberately does not privilege simulation. A test whose only trustworthy runner is a physical cell remains a first-class test; it is simply expensive, and its sampling budget must be allocated accordingly (§6.4).

4.3 Fidelity gating

A test earns the right to block a release; it does not hold that right by default, and realism is not the qualification—predictive validity is. Operationally: for each test–runner pair, estimate the rank correlation between its outcomes and real outcomes across a set of historical checkpoints of the same system; report it with every result; permit gating only above τ . This rule matters more in robotics than in software testing because a robotics test can look extremely convincing and still be uninformative—a photorealistic scene with guessed friction coefficients produces confident, wrong rankings, which is precisely the configuration current automated real-to-sim pipelines ship in [sim, 2026, rob, 2026c, sca, 2025].

Fidelity gating has a second effect that we consider its main practical justification: it makes low-fidelity tests *safe to keep*. A cable-routing suite with $\rho = 0.5$ still surfaces a large regression worth a real-world check; under a naive gate it would either block releases spuriously or be deleted. Advisory status retains the signal while denying it authority.

4.4 Evidence store

The store is append-only, addressed by release, and holds provenance sufficient to re-run each record. Three properties are load-bearing. **Immutability**, because an audit that can be edited is not an audit. **Release scoping**, so a query returns “what did we know when we shipped 4.2” rather than a current-state view. **Fidelity tagging**, so the gate rule is computable from the store rather than applied out of band. A machine-readable schema for the record of Definition 2 accompanies this paper.

4.5 Compliance mapping

Evidence produced for engineering reasons should discharge regulatory obligations without a second campaign. Table 2 gives the mapping we propose; the novel element is the treatment of model weights as SBOM components, which the CRA’s machine-readable SBOM requirement [a] does not contemplate and which no current tooling emits.

4.6 Cross-family inference

The composition claim is empirical and testable. We predict three interaction effects that no single-family evaluation can express.

1. **Robustness–competence trade-off.** A checkpoint that improves functional success may increase susceptibility to in-scene injection, if the gain came from stronger instruction-following. Given the per-model spread reported in [pap, 2026]—27.0%, 29.4%, 5.0%—variation of this magnitude across checkpoints of a single model family would be unsurprising and is, to our knowledge, unmeasured.
2. **Degradation-conditioned failure.** Functional failures should concentrate on units whose hardware degradation index exceeds a threshold. Stratifying functional results by hardware state converts unexplained cross-site variance into a maintenance action.
3. **Fidelity–family asymmetry.** A runner’s fidelity is family-dependent: a simulator adequate for adversarial scene perturbation may be inadequate for the contact physics of the same task. The framework records fidelity per pair precisely so this asymmetry is representable.

5 Metrics and proposed thresholds

Table 3 collects the metrics the framework requires. Thresholds marked *proposed* are ours and unvalidated; they are stated numerically so they can be falsified rather than debated.

6 Evaluation protocol for the framework

A framework of this kind is validated by whether its gate decisions would have been right. We specify four studies in increasing cost.

6.1 Retrospective gate study

Take a deployed fleet’s last N policy checkpoints, their real per-task outcomes, and their known field regressions. Construct suites without access to the outcomes, compute ρ per suite, apply the gate rule, and report rank correlation, regression catch rate, and false-block rate. This is the load-bearing experiment; it requires an industrial partner and no new capability.

6.2 Adversarial pre/post study

Instantiate the four attack categories of [pap, 2026] against a fixed policy, measure attack success, apply a guard layer, and re-measure. Report transfer from simulated to physical

presentation of the same attack, which is the quantity deciding whether simulated adversarial evidence may gate.

6.3 Cross-family interaction study

For a sequence of checkpoints, measure functional success and attack success jointly, and stratify functional failures by hardware degradation index. The predictions of §4.6 are either present or absent; either result is informative, and neither has been reported.

6.4 Budget allocation study

Given a fixed budget of physical-cell hours, compare allocation policies—uniform, fidelity-weighted, uncertainty-weighted—on regression catch rate. The relevant baseline is random sampling, which falsification work suggests is weak but which industrial practice currently approximates [rob, 2026b, for, 2026a].

7 Open problems

1. **Verifiers on contact-rich tasks.** The functional leg rests on success detection, which is near chance exactly where industrial value concentrates [fai, 2026]. We believe the missing evidence is non-visual—wrist force/torque, joint effort, vacuum pressure, tactile—and that verifier design should be multimodal rather than vision-only. This is a claim, not a result.
2. **Calibrating fidelity cheaply.** Estimating ρ requires historical checkpoints with real outcomes; new deployments have none. Transfer of fidelity estimates across sites and embodiments is unsolved.
3. **Physical parameters.** Every automated real-to-sim pipeline surveyed obtains mass and friction from a language model’s prior [sim, 2026, rob, 2026c, sca, 2025]. Identification from interaction logs is the obvious alternative and only partially explored [twi, 2025, asi, 2024].
4. **Compute budget of verification.** Verification-time scaling improves outcomes substantially but currently at control rates near 2.2 Hz [cov, 2026]; adaptive allocation is promising [ela, 2026] but trains per task.
5. **Standards coverage for learned components.** ISO 10218:2025 acknowledges ML functions but does not specify acceptance evidence for them; ISO 25785-1 is unpublished [, d]. That gap is where this framework is intended to be useful, and also why it cannot yet claim regulatory standing.
6. **Privacy as an evidence constraint.** First-person and workplace video is treated as biometric data under GDPR, with anonymisation expected before a dataset leaves the collection site. Evidence sharing between deployer, model provider, and assessor is therefore a data-protection design problem, not only a technical one.

8 Limitations

This paper proposes structure and measures nothing. Its thresholds are guesses with stated provenance. The status assignments in Table 1 are judgements formed from the surveyed

literature and would shift with a single strong result on contact-rich verification. Several industrial figures we cite—actuator fatigue as the binding constraint, end-effectors as the most failure-prone subsystem—come from trade and vendor sources rather than peer-reviewed work and should be read as practitioner consensus rather than measurement. The framework also assumes a release is a discrete event; continuous or per-site online adaptation breaks that assumption, and we do not address it.

9 Conclusion

Robotics does not lack tests. It lacks a rule for which tests are allowed to stop a release, and a place to keep the answer. The four families already exist in the literature; what the field has not done is compose them, tag each result with its measured predictive validity, and hold the composition against the release decision it is meant to inform. We have specified that composition precisely enough to be implemented and argued against, and named the one dependency that would make it work or fail: a verifier that can tell, on contact-rich tasks, whether the robot actually succeeded.

Reproducibility and artifacts

The project page, the LaTeX source of this paper, and a JSON Schema for the evidence record of Definition 2 are available in the accompanying repository. No datasets or trained models are released, as none were produced.

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Family	Tests	Primary metric	Status
Functional	Task success per family, cycle time, cross-task regression, field-failure replay	Success-rate delta vs. baseline, per task family, with interval	Gating for rigid tasks; advisory for deformable and insertion
Adversarial	In-scene prompt injection, patches, OOD scenes, sensor interference, trajectory redirection	Attack success rate per category, pre/post mitigation	Gating: attack success is directly observed, not inferred
Cyber	SBOM completeness incl. model artifacts, known-vulnerability exposure, update integrity, access paths	Unresolved exposure count by severity; time-to-evidence	Gating: deterministic, and externally mandated
Hardware	Actuator and transmission wear, end-effector degradation, calibration drift, thermal envelope	Degradation index and projected service interval	Advisory: coupling to task success unmodelled (§2.4)

Table 1: The four families. Status reflects the authors’ assessment against the surveyed literature, not a measured result.

Obligation	Instrument	Evidence that discharges it
Vulnerability and incident reporting	CRA, from 11 Sep 2026	Cyber-family exposure records with timestamps and disposition
Machine-readable SBOM	CRA, full 11 Dec 2027	Stack manifest extended with model artifacts: weights hash, training-data provenance class, licence
Security updates over support period	CRA	Release history with per-release gate outcome and update integrity test
ML functions altering behaviour over time	ISO 10218-1:2025	Functional-family regression record per release, plus adversarial-family results
Application and cell requirements	ISO 10218-2:2025	Site-scoped evidence bound to c in Definition 1
Dynamically stable mobile robots	ISO 25785-1 (draft)	Out of scope pending publication; structure anticipated

Table 2: Proposed mapping from engineering evidence to regulatory obligation. We produce evidence; we do not certify. Certification remains with accredited bodies, and vendor safety stacks occupy the functional-safety layer beneath this table.

Metric	Threshold	Basis
Suite fidelity ρ : rank correlation between suite and real outcomes over ≥ 8 historical checkpoints	≥ 0.80	Proposed; anchored to 0.887–0.911 [sim, 2026, rob, 2026c]
Rank violation: maximum inversion between suite and real ranking	≤ 0.05	Proposed; 0.018 in [sim, 2026], 0.0066 in [rob, 2026c]
Verifier balanced accuracy, reported separately for contact-light and contact-heavy tasks	≥ 0.90	Proposed; best published 0.77 overall, < 0.60 contact-intensive [fai, 2026]
Attack success rate per category, pre/post mitigation	report	Measured directly; protocol from [pap, 2026]
Regression catch rate on known past field regressions	≥ 0.80	Proposed; §6.1 would establish it
Time-to-evidence: checkpoint availability to complete gate decision	≤ 24 h	Proposed; set by release cadence
Test-time compute cost as multiple of base inference	report	Best-of- N at ≈ 2.2 Hz [cov, 2026]; adaptive matches best-of-10 at 34% lower latency [ela, 2026]

Table 3: Metrics, with thresholds stated as falsifiable proposals rather than findings.